AI-Assisted Low-Cost Sensing for Everyday Infrastructure: A Smartphone-Based Elevator Dynamics Case Study

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Abstract

Modern infrastructure depends on sensing, but many sensing workflows remain expensive, expertise-intensive, and difficult for ordinary users to repeat. This creates a practical gap between systems that people experience every day and the quantitative measurements needed to understand their performance. Here I describe an AI-assisted low-cost sensing workflow that combines human problem selection, smartphone data acquisition, agent-assisted signal analysis, and reusable workflow packaging. As a feasibility case, I measured the vertical motion of three campus elevators using smartphone linear-acceleration recordings collected at approximately 200 Hz. The phone was placed on the elevator floor, and the resulting acceleration records were analyzed to obtain relative velocity, jerk, frequency spectra, autocorrelation structure, and comparative summary indicators. The case study shows that the three elevators produced distinct ride signatures. The Science Teaching Building elevator had the shortest detected motion time at 55.02 s, whereas the Laboratory elevator had the lowest RMS jerk at 0.183 m/s^3, indicating gentler acceleration transitions. Spectral energy in the 1.5–2.5 Hz walking-cadence band was small, so the data do not support a strong resonance claim. These results demonstrate that smartphone sensing and AI-assisted analysis can support preliminary, low-cost characterization of everyday infrastructure, while remaining clearly distinct from certified safety inspection.

1 Introduction

Modern life depends on infrastructure that is often noticed only when it fails or feels uncomfortable. Power systems, bridges, railways, laboratory buildings, dormitories, and elevators are ordinary parts of daily life, but each contains mechanical, electrical, and structural behavior that can be measured. Sensing such systems matters because measurements connect subjective experience with evidence. A person may say that one elevator feels faster, another feels smoother, or a third vibrates more, but these statements remain impressions until they are connected to acceleration, vibration, and other physical quantities.

Professional sensing methods are powerful but not always accessible. In industrial or safety-critical contexts, measurement normally requires calibrated instruments, trained personnel, specialized workflows, and formal standards. Such requirements are appropriate when the aim is certification or safety inspection. However, they also leave a gap for educational and preliminary investigation. A student, building user, or ordinary citizen may be able to observe a problem but may not have the tools or expertise to translate that observation into quantitative evidence. This gap is especially important for scientific writing and communication because everyday systems can become useful scientific examples only when the measurement logic is clear and reproducible.

Smartphones provide one way to reduce this barrier. Modern phones contain accelerometers, gyroscopes, pressure sensors, microphones, cameras, and time-stamped data-recording software.

Smartphone-based experiments have already been used in physics education and low-cost measurement because the device is familiar, portable, and sufficiently sensitive for many preliminary studies [1]. Previous work has also shown that smartphone sensors can be used to study vertical motion in elevators and related systems, while also warning that numerical integration of accelerometer data requires caution because sensor noise and bias accumulate over time [2]. These studies suggest that consumer devices can serve as practical scientific instruments when the measurement question is matched to the device's limitations.

AI agents add a second opportunity: they can help connect a vague real-world question to a structured measurement workflow. In a human-AI collaborative process, the human chooses the problem, judges whether the result is meaningful, and checks whether the claim is appropriate. The AI agent can help propose measurable quantities, organize signal-processing steps, generate analysis code, compare possible interpretations, and package the workflow for later reuse. Recent discussions of large-language-model agents describe such systems as tools that can plan, use tools, and support multi-step work, although they still require human oversight and careful verification [6, 7]. For a small undergraduate study, this means that the agent is not a replacement for scientific judgment. Instead, it can act as a practical assistant for turning an everyday observation into a coherent experiment.

The research gap addressed here is therefore not that elevators have never been studied. Elevators are engineered systems, and ride comfort and vibration exposure are already connected to established measurement frameworks such as ISO 2631-1 for whole-body vibration [3]. The gap is more practical and educational: there are fewer demonstrations of a complete, low-cost, AI-assisted workflow that begins with an ordinary infrastructure question and ends with interpretable quantitative results that a non-specialist can understand. Such a workflow should not pretend to replace professional diagnostics. It should instead show how a user can make a careful first measurement, avoid overclaiming, and communicate what the data do and do not support.

This paper presents that workflow through a smartphone-based elevator case study. I measured three campus elevators and used AI-assisted analysis to process acceleration records into ride signatures. The study asks whether a phone plus an AI-assisted analysis pipeline can distinguish differences in elevator motion, identify interpretable comfort-related indicators, and produce a reusable structure for similar low-cost sensing tasks. The contribution is twofold: first, the paper demonstrates a general workflow for accessible sensing; second, it provides a concrete elevator example in which the claims are limited to preliminary motion characterization.

2 Methods

The study used a human-AI collaborative workflow with five stages. First, the human selected an everyday infrastructure problem: different campus elevators feel different in speed and smoothness. Second, the AI agent helped translate this problem into measurable quantities, including acceleration, relative velocity, jerk, vibration spectrum, and summary ride indicators. Third, the human collected smartphone sensor data from three elevators. Fourth, the agent-assisted analysis pipeline processed the data and produced figures and metrics. Fifth, the workflow was treated as reusable: the same structure could later be applied to other facilities, repeated trials, or additional sensors.

The experimental case used three elevators on campus: the Laboratory elevator, the Science Teaching Building elevator, and the Second Teaching Building elevator. For each elevator, one representative ride was recorded using a smartphone sensor application capable of exporting linear acceleration as a CSV file. The phone was placed on the floor of the elevator rather than held in

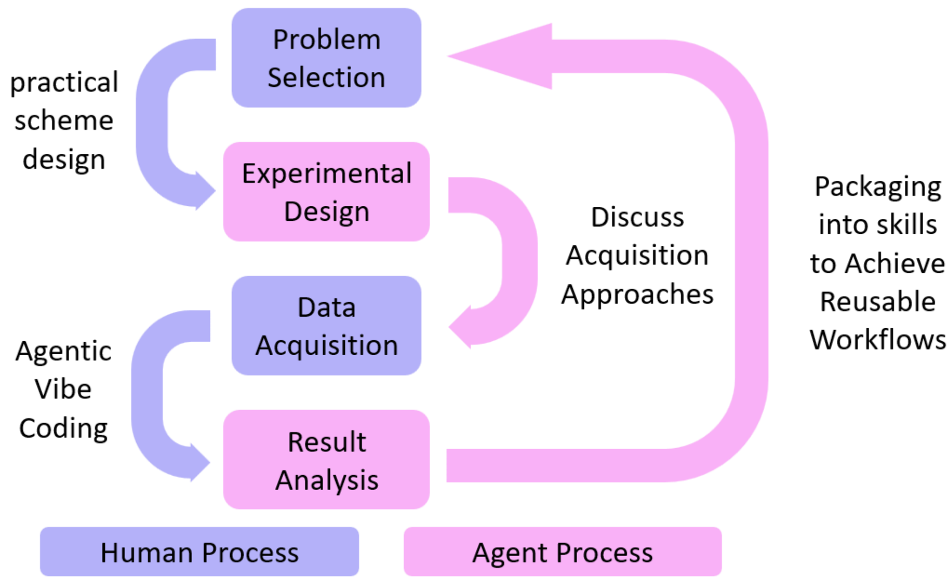

Figure 1: AI-assisted low-cost sensing workflow. The workflow begins with human problem selection and practical scheme design, moves through agent-supported experimental design and data-acquisition planning, and then uses agentic analysis to process results. The loop returns to the human for discussion of acquisition approaches and interpretation before packaging the procedure into reusable skills. Purple elements indicate human-led stages, and pink elements indicate agent-assisted stages.

the hand, so that hand tremor and body motion would not dominate the acceleration signal. Each CSV file contained time, linear acceleration along the x, y, and z axes, and absolute acceleration. The effective sampling frequencies estimated from the time stamps were approximately 201 Hz for all three records.

The primary physical signal was the vertical linear acceleration, denoted here as $a_z(t)$. This choice follows the fact that elevator motion is mainly vertical and that ride transitions are most directly visible in the vertical acceleration record. The absolute acceleration was also retained as a complementary quantity because it captures total sensor motion and can reveal non-vertical vibration. The records were not treated as certified measurements of elevator performance. They were treated as consumer-grade measurements suitable for comparing motion signatures within this small dataset.

Before deriving secondary quantities, the acceleration records were standardized and smoothed. Smartphone accelerometers contain high-frequency noise, and numerical differentiation can amplify that noise. Therefore, smoothing was applied before calculating jerk, following the general logic of least-squares smoothing and differentiation methods such as the Savitzky-Golay approach [4]. The purpose of smoothing was not to hide meaningful motion but to make the derived jerk and spectral indicators interpretable. The method was kept conservative: the important acceleration transitions remained visible in the time-domain plots.

Relative velocity was reconstructed by numerical integration of the vertical acceleration. This quantity was used as a motion signature, not as a calibrated elevator speed. Accelerometer integration is sensitive to bias and drift, and this limitation is well known in smartphone motion measurements [2]. For this reason, the velocity plots in this paper are interpreted comparatively.

They show how the three rides differ in acceleration timing and transition structure, but they are not used to certify absolute elevator speed.

Jerk was calculated as the time derivative of vertical acceleration. In ride-comfort analysis, jerk is useful because it describes how abruptly acceleration changes. A ride may have modest acceleration but still feel uncomfortable if acceleration changes sharply. In this study, RMS jerk was used as one indicator of smoothness: a lower RMS jerk suggests gentler transitions during the representative ride. Peak jerk was also calculated, but RMS jerk was emphasized because it describes the overall transition roughness across the motion window rather than a single extreme point.

Frequency-domain analysis was used to examine vibration content. The acceleration signal was transformed into a power spectral density representation using Fourier-based methods, following the general principle of the fast Fourier transform [5]. The analysis also marked the 1.5–2.5 Hz walking-cadence band as a perceptual reference. This band was not used to prove resonance. Instead, it provided a simple way to ask whether a large amount of elevator vibration energy overlapped with a frequency range that people commonly experience during rhythmic walking.

Autocorrelation was used to examine repeated or oscillatory structure in the acceleration during the motion stage. If a signal contains regular vibration, its autocorrelation can show repeated peaks at characteristic time lags. In this study, autocorrelation was used as a descriptive tool rather than as a formal model of elevator mechanics. It helped compare whether the three rides showed different decay or repeated-motion patterns after major acceleration transitions.

Motion-stage detection was used to define comparable analysis windows. For each record, the processed acceleration trace was used to identify the interval containing the main acceleration, travel, and deceleration behavior, while excluding the longer waiting period before motion when possible. The stage-detection status was “ok” for all three records in the summary table. This procedure produced the reported total motion times and also defined the residual-vibration interval after arrival. Because the method used one representative recording per elevator, these times describe the measured rides rather than fixed properties of the elevators.

Finally, several summary indicators were calculated from each record. These included total detected motion time, RMS acceleration, peak acceleration, peak and RMS jerk, residual vibration RMS after arrival, dominant frequency, low-frequency energy, walking-band energy, autocorrelation peak lag, and autocorrelation decay time. A radar chart converted selected indicators into relative within-dataset scores: efficiency, low acceleration, low jerk, low residual vibration, and walking-band avoidance. These scores compare only the three records in this study. They should not be interpreted as universal elevator-quality grades.

3 Results

The smartphone recordings produced distinct acceleration signatures for the three elevators (Figure 2a). In the representative time-domain acceleration plots, each ride contained visible acceleration and deceleration stages rather than a flat or featureless signal. The Laboratory record showed several separated acceleration events and a strong residual component near the end of the displayed segment. The Science Teaching Building record showed a shorter overall motion pattern with a clear acceleration-deceleration structure. The Second Teaching Building record showed the longest detected motion interval and a broader acceleration pattern. These differences support the basic feasibility claim: a phone placed on the elevator floor can capture ride-specific motion signatures.

Velocity reconstruction (Figure 2b) clarified the timing of the motion stages. Because velocity

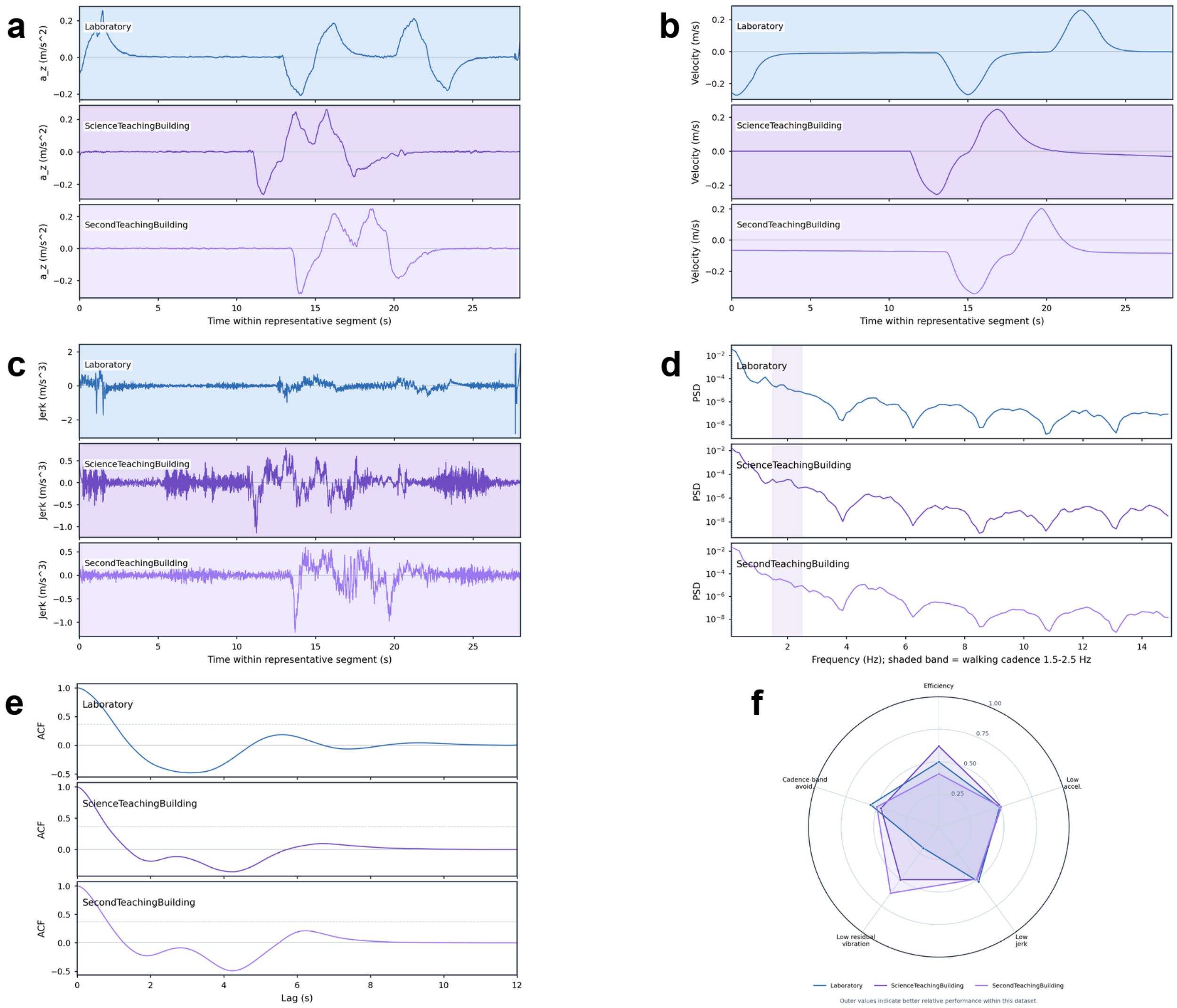

Figure 2: Combined smartphone elevator dynamics results. (a) Representative vertical acceleration profiles show distinct acceleration and deceleration structures for the Laboratory, Science Teaching Building, and Second Teaching Building elevators. (b) Relative velocity profiles reconstructed from acceleration summarize motion timing but are interpreted only as relative signatures because accelerometer integration can drift. (c) Jerk profiles quantify transition abruptness; the Laboratory ride had the lowest RMS jerk, indicating gentler transitions by this metric. (d) Frequency-domain spectra show low walking-cadence-band energy in the shaded 1.5–2.5 Hz band, so strong resonance was not observed. (e) Autocorrelation profiles compare repeated temporal structure without assigning a specific mechanical cause. (f) Normalized quality scores compare the three measured records within this dataset only; outer values indicate better relative performance, not universal elevator-quality grades.

was obtained by integrating acceleration, it is best read as a relative profile rather than an absolute speed record. The Science Teaching Building elevator had the shortest detected motion time, 55.02 s, compared with 89.97 s for the Laboratory elevator and 130.49 s for the Second Teaching Building elevator. This result supports the practical statement made in the presentation: among the three measured cases, the Science Teaching Building elevator was the fastest according to detected motion time. The conclusion is local to the measured rides, because travel distance and operating conditions were not independently standardized.

Jerk analysis (Figure 2c) separated smoothness from speed. The Laboratory elevator had the lowest RMS jerk, 0.183 m/s^3, whereas the Science Teaching Building and Second Teaching Building elevators had RMS jerk values of 0.198 m/s^3 and 0.202 m/s^3, respectively. This means that the Laboratory ride had gentler acceleration transitions according to this metric. However, the same record also had the largest residual vibration RMS after arrival, 0.519 m/s^2. Therefore, the correct conclusion is not that the Laboratory elevator was the best overall. The narrower and better-supported conclusion is that the Laboratory ride had the lowest RMS jerk in this dataset, while other indicators reveal different tradeoffs.

Frequency spectra (Figure 2d) showed dominant low-frequency motion near 0.125 Hz in all three records. The low dominant frequency is consistent with the overall ride cycle rather than a high-frequency vibration mode. Energy in the 1.5–2.5 Hz walking-cadence band was small for all three rides: approximately 0.000016 for Laboratory, 0.000022 for Science Teaching Building, and 0.000019 for Second Teaching Building in the summary table. Thus, the analysis does not support a strong resonance claim with walking cadence. This negative result is important because it prevents the paper from overstating the meaning of the frequency plot.

Autocorrelation (Figure 2e) provided an additional view of repeated motion structure. The three records differed in their autocorrelation decay and peak-lag patterns, with peak lags of approximately 5.59 s for Laboratory, 6.70 s for Science Teaching Building, and 6.23 s for Second Teaching Building. These values should be interpreted descriptively. They show that the acceleration records contain different temporal structures after preprocessing, but they do not by themselves identify a mechanical source. In a professional diagnostic setting, repeated measurements and independent mechanical information would be required before assigning the pattern to a specific elevator component.

The summary indicators (Figure 2f) reveal why a single word such as “best” is not sufficient. The Science Teaching Building elevator scored well on efficiency because it had the shortest detected motion time. The Laboratory elevator scored well on low jerk but poorly on residual vibration. The Second Teaching Building elevator had the lowest residual vibration RMS among the three records but the longest detected motion time. The radar chart therefore communicates a comparative profile rather than a ranking. It shows the value of multi-metric sensing: different quantities answer different user questions.

Overall, the elevator case supports the proposed workflow. The smartphone collected usable motion data, the analysis pipeline transformed the records into interpretable indicators, and the results produced cautious conclusions that match the data. The workflow also ruled out an over-strong claim: although a walking-cadence band was inspected, strong resonance was not observed. This combination of positive findings and restrained interpretation is central to low-cost sensing. A preliminary tool is useful not because it says everything, but because it helps users ask better questions with evidence.

4 Discussion

The main result of this study is that AI-assisted low-cost sensing can turn an everyday infrastructure experience into a quantitative investigation. The phone provided the measurement, but the workflow provided the scientific structure. Without the workflow, the raw CSV files would remain difficult to interpret. Without the phone, the workflow would have no data. Without human judgment, the AI agent could easily overstate the meaning of the figures. The value of the approach lies in combining these roles: human problem choice, accessible sensing, agent-assisted analysis, and cautious interpretation.

The elevator case also shows why the research question should be framed around workflow rather than novelty in elevator engineering. A professional engineer would not use one smartphone ride recording as a safety certification. That is not the claim of this paper. Instead, the claim is that a student or ordinary user can perform a low-cost preliminary comparison that identifies measurable differences and motivates better follow-up questions. For example, if an elevator shows unusually high jerk or residual vibration in repeated recordings, that result could justify more careful inspection by qualified personnel. The smartphone workflow therefore functions as an entry point into measurement, not as a replacement for professional systems.

AI assistance was useful mainly because the task crossed several small domains. The study required physical reasoning about acceleration and jerk, practical decisions about phone placement, data processing for time series, figure design, and scientific writing discipline. An AI agent can help connect these pieces by suggesting metrics, writing analysis code, checking whether results answer the original question, and turning a one-time script into a reusable skill. However, this assistance also creates risk. The agent may suggest an impressive interpretation that the data do not support. The safest workflow is therefore collaborative: the AI proposes and organizes, while the human verifies, limits claims, and checks whether the result remains physically plausible.

The findings should be interpreted through the limitations of consumer-grade sensing. The smartphone was not calibrated against a professional accelerometer, and each elevator was represented by one recording. Travel distance, passenger load, direction of motion, elevator scheduling behavior, phone orientation, and building conditions were not systematically controlled. The numerical velocity profiles are relative because acceleration integration can drift. The radar scores are relative to this small dataset. These limitations prevent broad claims about the elevators themselves, but they do not invalidate the workflow demonstration. A feasibility study can be successful if it shows that the method produces meaningful, cautious, and reproducible first-order evidence.

The method can be strengthened in several straightforward ways. Future work should record repeated rides for each elevator, including upward and downward trips, different passenger loads, and different times of day. A fixed phone holder could reduce orientation variation. A second phone or a calibrated reference sensor could estimate measurement uncertainty. The analysis could report confidence intervals rather than single values. The workflow could also be packaged into a reusable template: collect CSV data, check sampling rate, detect motion stages, compute acceleration and jerk metrics, generate figures, and produce a short interpretation report. This would make the AI-assisted workflow more useful for other everyday sensing tasks.

The broader implication is that low-cost sensing can improve scientific literacy. Many students learn scientific method through laboratory apparatus that feels separate from daily life. Smartphone sensing reverses the direction: it begins from a familiar object and a familiar environment, then asks how evidence can be extracted. The elevator example is useful because most readers know what an elevator feels like, but not all readers have thought about acceleration, jerk, and vibration as measurable features of that experience. A clear workflow can therefore make mechanics more tangible and make data analysis more purposeful.

At the same time, accessible sensing requires careful communication. The easier it becomes to generate figures, the easier it becomes to overinterpret them. This is why the paper repeatedly distinguishes preliminary assessment from certified inspection. It is also why the negative result about walking-cadence resonance matters. A strong paper does not need every result to be dramatic. It needs the conclusions to be supported by the data. In this case, the data support three modest conclusions: the measured elevators have distinct motion signatures; the Science Teaching Building ride was shortest in detected motion time; and the Laboratory ride had the lowest RMS jerk in the dataset. These conclusions are narrower than a safety diagnosis, but they are scientifically meaningful.

5 Conclusion

This paper presented an AI-assisted low-cost sensing workflow and verified it with a smartphone-based elevator dynamics case study. The workflow begins with a human-chosen infrastructure question, uses a smartphone to collect motion data, applies agent-assisted signal analysis, and produces interpretable results that can be reused or extended. In the elevator case, smartphone acceleration records distinguished three campus rides and supported specific conclusions about motion time, jerk, vibration, and frequency content.

The key achievement is not a new theory of elevator dynamics. The achievement is a practical demonstration of accessible measurement. The Science Teaching Building elevator had the shortest detected motion time, the Laboratory elevator had the lowest RMS jerk, and the frequency analysis did not show strong walking-band resonance. These findings show how everyday experience can be converted into quantitative evidence while preserving appropriate caution.

Future work should repeat measurements, control ride conditions, estimate uncertainty, and package the analysis pipeline into a reusable AI-assisted sensing skill. With these improvements, similar workflows could be applied to other campus systems, including doors, stairs, bicycles, laboratory equipment, or building vibrations. The take-home message is simple: smartphones and AI agents can help ordinary users begin measuring the engineered world around them, provided the resulting claims remain as careful as the measurements that support them.

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